

UNIVERSITY OF BRISTOL

Faculty of Science

MOCK EXAMINATION — Paper 8 (Extreme)
Core Physics I — Classical, Quantum and Thermal Physics
Module Code: PHYS10012

Duration: 3 Hours

Total Marks: 100

Instructions:

- Answer **ALL** questions.
 - **Section A** (Oscillations and Waves): 10 multiple-choice questions, 20 marks total.
 - **Section B** (Quantum Physics): 10 multiple-choice questions, 20 marks total.
 - **Section C**: 4 long-answer questions, 60 marks total.
 - An approved calculator is permitted.
 - A formula sheet is provided at the end of this paper.
 - Show all working for Section C. Marks will be awarded for clear, logical solutions.
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Section A: Oscillations and Waves

Answer all 10 questions. Each question is worth 2 marks.

1. A particle of mass m moves in the potential $V(x) = V_0 \left(1 - \cos\left(\frac{x}{a}\right)\right)$, where V_0 and a are positive constants. For small displacements about $x = 0$, what is the angular frequency ω of oscillation? [2 marks]

- A. $\sqrt{\frac{V_0}{ma^2}}$
- B. $\sqrt{\frac{2V_0}{ma^2}}$
- C. $\sqrt{\frac{V_0}{2ma^2}}$
- D. $\sqrt{\frac{V_0}{ma}}$

2. A damped harmonic oscillator has damping constant γ and initial energy E_0 . The amplitude decays as $A(t) = A_0 e^{-\gamma t/2}$. Which expression gives the energy $E(t)$ at time t ? [2 marks]

- A. $E_0 e^{-\gamma t/2}$
- B. $E_0 e^{-\gamma t}$
- C. $E_0 e^{-2\gamma t}$
- D. $E_0 (1 - e^{-\gamma t})$

3. For a lightly damped forced oscillator with damping constant γ , the full-width at half-maximum (FWHM) of the *power absorption* curve (plotted against driving frequency ω) is approximately: [2 marks]

- A. $\gamma/2$
- B. γ
- C. 2γ
- D. ω_0/Q

4. Two harmonic waves of the same frequency and wavenumber travel in opposite directions on a string: $y_1 = A_1 \sin(kx - \omega t)$ and $y_2 = A_2 \sin(kx + \omega t)$, where $A_1 \neq A_2$. The superposition $y_1 + y_2$ is: [2 marks]

- A. A pure standing wave.
- B. A pure travelling wave.

- C. A standing wave superimposed with a travelling wave.
- D. Zero everywhere.

5. Three strings under the same tension are joined in series: string 1 (impedance Z_1), a middle section of length ℓ (impedance Z_m), and string 3 (impedance Z_3). If the middle section has length $\ell = \lambda/4$ (a quarter-wavelength at the operating frequency), what value of Z_m eliminates all reflection from an incident wave in string 1? [2 marks]

- A. $Z_m = (Z_1 + Z_3)/2$
- B. $Z_m = Z_1 Z_3 / (Z_1 + Z_3)$
- C. $Z_m = \sqrt{Z_1 Z_3}$
- D. $Z_m = Z_3$

6. Two identical masses m are connected by springs of constant k to fixed walls, and by a coupling spring k_c to each other. In the matrix equation $m\ddot{\mathbf{x}} = -\mathbf{K}\mathbf{x}$ (where $\mathbf{x} = (x_1, x_2)^T$), which matrix $\mathbf{K}$ gives the correct equations of motion? [2 marks]

- A. $\begin{pmatrix} k + k_c & -k_c \\ -k_c & k + k_c \end{pmatrix}$
- B. $\begin{pmatrix} k & k_c \\ k_c & k \end{pmatrix}$
- C. $\begin{pmatrix} k + k_c & k_c \\ k_c & k + k_c \end{pmatrix}$
- D. $\begin{pmatrix} k & -k_c \\ -k_c & k \end{pmatrix}$

7. A wave packet propagates in a medium with dispersion relation $\omega(k)$. The wave packet will spread (disperse) over time if: [2 marks]

- A. $\frac{d\omega}{dk} \neq 0$
- B. $\frac{d^2\omega}{dk^2} \neq 0$
- C. $\frac{d\omega}{dk} = 0$
- D. ω/k is constant for all k

8. A source emitting frequency f_0 moves toward a stationary observer at speed $v_s \ll c$. The classical Doppler formula gives $f'_{\text{cl}} = f_0/(1 - v_s/c)$ and the relativistic formula gives $f'_{\text{rel}} = f_0 \sqrt{(1 + \beta)/(1 - \beta)}$ where $\beta = v_s/c$. To first order in β , the relativistic correction modifies the classical result by a factor of approximately: [2 marks]

- A. $1 - \beta/2$

B. $1 + \beta/2$

C. $1 + \beta$

D. There is no first-order correction; they agree to first order in β .

9. A standing wave $y = 2A \sin(kx) \cos(\omega t)$ is formed on a string of length $L = n\lambda/2$ (i.e. n half-wavelengths fit). How does its total energy compare to that of a single travelling wave $y = A \sin(kx - \omega t)$ on the same string? [2 marks]

A. The standing wave has twice the total energy.

B. The standing wave has the same total energy.

C. The standing wave has half the total energy.

D. The energies cannot be compared without knowing n .

10. An object moves through air at Mach 2 (i.e. at twice the speed of sound). The half-angle θ of the Mach cone (shock wave cone) is: [2 marks]

A. 60°

B. 45°

C. 30°

D. 15°

Section B: Quantum Physics

Answer all 10 questions. Each question is worth 2 marks.

1. If $[\hat{A}, \hat{B}] = i\hat{C}$ for operators $\hat{A}$, $\hat{B}$, and $\hat{C}$, what is $[\hat{B}, \hat{A}]$? [2 marks]
 - A. $i\hat{C}$
 - B. $-i\hat{C}$
 - C. $\hat{C}$
 - D. 0
2. If two Hermitian operators $\hat{A}$ and $\hat{B}$ satisfy $[\hat{A}, \hat{B}] = 0$, which statement is correct? [2 marks]
 - A. Measuring $\hat{A}$ always gives the same result as measuring $\hat{B}$.
 - B. $\hat{A}$ and $\hat{B}$ share a common set of eigenstates, and both can be measured simultaneously without mutual disturbance.
 - C. $\hat{A}$ and $\hat{B}$ must be equal.
 - D. The uncertainty product $\Delta A \Delta B$ must equal $\hbar/2$.
3. Let U be the unitary matrix whose columns are the normalised eigenvectors of a Hamiltonian H . Then $U^\dagger H U$ equals: [2 marks]
 - A. The identity matrix I .
 - B. H itself.
 - C. A diagonal matrix with the eigenvalues of H on the diagonal.
 - D. The zero matrix.
4. The no-cloning theorem states that it is impossible to create an exact copy of an arbitrary unknown quantum state. This is fundamentally because: [2 marks]
 - A. Copying requires infinite energy.
 - B. The Heisenberg uncertainty principle forbids knowing the state precisely.
 - C. Quantum evolution is linear (unitary), and linearity is incompatible with a universal cloning operation.
 - D. Entanglement prevents the state from being localised.
5. For a system in a pure state $|\psi\rangle$, the density matrix is $\rho = |\psi\rangle\langle\psi|$. What is $\text{Tr}(\rho^2)$? [2 marks]
 - A. 0

- B. $1/2$
- C. 1
- D. It depends on the state $|\psi\rangle$.

6. Two energy eigenstates of a system have energies E_1 and E_2 . A superposition of these states will exhibit oscillatory behaviour. The characteristic timescale τ of these oscillations is: [2 marks]

- A. $\tau = \hbar/(E_1 + E_2)$
- B. $\tau = \hbar/|E_1 - E_2|$
- C. $\tau = h/(E_1 E_2)$
- D. $\tau = (E_1 - E_2)/\hbar$

7. Two spin- $\frac{1}{2}$ particles are in the singlet state $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$. What is the expectation value $\langle S_{1z} S_{2z} \rangle$? [2 marks]

- A. $+\hbar^2/4$
- B. $-\hbar^2/4$
- C. 0
- D. $-\hbar^2/2$

8. Two qubits are in the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)$. The reduced density matrix ρ_1 of qubit 1, obtained by tracing over qubit 2, is: [2 marks]

- A. $|\uparrow\rangle\langle\uparrow|$
- B. $\frac{1}{2}|\uparrow\rangle\langle\uparrow| + \frac{1}{2}|\downarrow\rangle\langle\downarrow|$
- C. $\frac{1}{2}(|\uparrow\rangle\langle\uparrow| + |\uparrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|)$
- D. $\frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)\frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$

9. N non-interacting spin- $\frac{1}{2}$ fermions are placed in a one-dimensional infinite square well of width L . Each single-particle energy level $E_n = n^2 E_1$ (where $E_1 = \pi^2 \hbar^2 / (2mL^2)$) can hold at most 2 fermions (one spin-up, one spin-down). The ground-state energy of the system for $N = 6$ fermions is: [2 marks]

- A. $6E_1$
- B. $14E_1$
- C. $28E_1$
- D. $42E_1$

10. Which discrete symmetry is violated in weak interactions but preserved in electromagnetic and strong interactions? [2 marks]

- A. Charge conjugation (C) only.
- B. Parity (P).
- C. Time reversal (T) only.
- D. The combined symmetry CPT .

Section C: Long Answer Questions

Answer all 4 questions. Show all working.

Question 1.

[15 marks]

Snooker Ball Impulse, Rolling Friction, and Loop-the-Loop

- (a) A uniform solid sphere (billiard ball) of mass M and radius R rests on a horizontal surface with coefficient of kinetic friction μ_k . A cue strikes the ball with a horizontal impulse J at a height h above the centre of the ball.
- (i) Find the initial translational velocity v_0 and the initial angular velocity ω_0 of the ball immediately after the impulse. 2
 - (ii) Show that if the cue strikes at height $h = \frac{2}{5}R$ above the centre, the ball immediately rolls without slipping (i.e. $v_0 = R\omega_0$), so no friction is needed. At what height above the *table* (i.e. above the lowest point of the ball) should the cue strike? 3
- (b) For a general strike height $h \neq \frac{2}{5}R$, the ball initially slides. Friction acts until rolling is achieved.
- (i) Write the equations of motion for the translational and rotational motion during the sliding phase. 2
 - (ii) Find the time t_{roll} at which rolling without slipping begins and the velocity v_{roll} at that instant. For a centre-strike ($h = 0$), show that $v_{\text{roll}} = \frac{5}{7}v_0$. 3
- (c) A solid disk of mass $M = 0.5$ kg and radius $R = 0.15$ m is released from rest at height H (measured from the bottom of a circular loop) on an inclined track. At the bottom of the incline the track forms a vertical circular loop of radius $r = 0.8$ m. The disk rolls without slipping throughout.
- Find the minimum height H for the disk to maintain contact with the track at the top of the loop. 5

Question 2.

[15 marks]

Advanced Entropy, Lennard-Jones Potential, and Maxwell–Boltzmann Distribution

- (a) Two identical blocks of the same material (mass m , specific heat capacity c) are initially at temperatures T_1 and T_2 ($T_1 > T_2$). A series of Carnot engines operate between them, each extracting an infinitesimal amount of work, until the blocks reach the same final temperature T_f .
- Show that $T_f = \sqrt{T_1 T_2}$. 2
 - Show that the total entropy change of the two blocks is zero. 2
- (b) If instead the two blocks are simply placed in direct thermal contact (an irreversible process):
- Show that the final temperature is $T_f' = \frac{T_1 + T_2}{2}$. 1
 - Calculate the total entropy change ΔS_{total} and show that it is positive. 2
 - Hence show that the maximum extractable work from the Carnot process in part (a) was $W_{\text{max}} = mc(T_1 + T_2 - 2\sqrt{T_1 T_2})$. 1
- (c) The Lennard-Jones potential is $V(r) = 4\epsilon \left[\left(\frac{a}{r} \right)^{12} - \left(\frac{a}{r} \right)^6 \right]$.
- Show that the equilibrium separation is $r_0 = 2^{1/6} a$ and that the depth of the potential well is ϵ . 2
 - By expanding $V(r)$ about r_0 to second order, show that the effective spring constant is $k = 72 \epsilon / (2^{1/3} a^2)$. Hence estimate the frequency of small oscillations of a single argon atom about its equilibrium position, using $\epsilon = 1.67 \times 10^{-21}$ J, $a = 3.40 \times 10^{-10}$ m, and $m_{\text{Ar}} = 6.63 \times 10^{-26}$ kg. 2
- (d) The most probable speed in the Maxwell–Boltzmann distribution is $c_{\text{mp}} = \sqrt{2k_B T/m}$. At what temperature does the most probable speed of helium atoms ($m_{\text{He}} = 6.64 \times 10^{-27}$ kg) equal the escape velocity from the Earth's surface ($v_e = 11.2$ km/s)? Comment on whether helium can escape from the Earth's atmosphere at room temperature. 3

Question 3.

[15 marks]

Advanced Boundary Conditions, Beaded String, and Wave Packets

- (a) A string of length L is fixed at $x = 0$ and attached to a massless ring that slides freely (without friction) on a vertical rod at $x = L$.
- What is the boundary condition on the displacement $y(x, t)$ at $x = L$? 1
 - Show that the allowed wavelengths are $\lambda_n = \frac{4L}{2n-1}$ for $n = 1, 2, 3, \dots$ 2
 - If $L = 2$ m and $v = 100$ m/s, find the first three resonant frequencies. 2
- (b) Two identical strings (same linear mass density μ , same tension T) of length L are connected at $x = 0$ by a bead of mass m . The left string is fixed at $x = -L$ and the right string is fixed at $x = +L$.
- Write down the boundary conditions at $x = \pm L$ and the matching conditions at $x = 0$ (continuity of displacement and the force balance including the bead's inertia). 3
 - For the *symmetric* normal modes, the bead sits at a node (i.e. $y(0, t) = 0$). Show that in this case the resonant frequencies are $f_n = nv/(2L)$ for $n = 1, 2, 3, \dots$ 2
- (c) A wave packet is formed from two harmonic waves:

$$y(x, t) = A \cos(k_1 x - \omega_1 t) + A \cos(k_2 x - \omega_2 t),$$

where $k_1 = 10$ rad/m, $k_2 = 12$ rad/m, and the dispersion relation is $\omega = \alpha k^2$ with $\alpha = 0.5$ m²/s.

- Calculate ω_1 and ω_2 . 1
- Find the phase velocity v_p and group velocity v_g at the average wavenumber $\bar{k} = (k_1 + k_2)/2$. 2
- Rewrite $y(x, t)$ as a product of a slowly-varying envelope and a rapidly-varying carrier. Identify which factor moves at the group velocity and which at the phase velocity. 2

Question 4.

[15 marks]

Three-State System, Fermion States, and Pion Decay

- (a) Consider a three-state quantum system with orthonormal basis $\{|a\rangle, |b\rangle, |c\rangle\}$. The Hamiltonian is

$$H = E_0(|a\rangle\langle a| + |b\rangle\langle b| + |c\rangle\langle c|) + V(|a\rangle\langle b| + |b\rangle\langle a|),$$

where E_0 and V are real constants.

- (i) Write H as a 3×3 matrix in the $\{|a\rangle, |b\rangle, |c\rangle\}$ basis. 1
 - (ii) Find all three eigenvalues and the corresponding normalised eigenstates. 3
- (b) At $t = 0$ the system is prepared in the state $|\psi(0)\rangle = |a\rangle$.
- (i) Find $|\psi(t)\rangle$. 2
 - (ii) Calculate the probability $P(a, t) = |\langle a|\psi(t)\rangle|^2$ of finding the system in state $|a\rangle$ at time t . 2
- (c) Two identical spin- $\frac{1}{2}$ fermions occupy a system with two spatial orbitals $|\alpha\rangle$ and $|\beta\rangle$. Write down *all* possible two-particle states that are antisymmetric under exchange of the two fermions (combining spatial and spin degrees of freedom). State the total number of such states. 3
- (d) Consider the decay $\pi^0 \rightarrow \gamma + \gamma$ (neutral pion decaying into two photons).
- (i) Verify that charge, baryon number, and lepton number are all conserved in this decay. 2
 - (ii) The π^0 has spin 0. Using conservation of angular momentum, discuss the constraint on the helicities (circular polarisation states) of the two emitted photons. Which force mediates this decay? 2

Formula Sheet

Mathematical Formulae

Binomial expansion: $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (|x| \ll 1)$

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

Trig identities: $\sin A + \sin B = 2 \cos\left(\frac{A-B}{2}\right) \sin\left(\frac{A+B}{2}\right)$
 $\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

Mechanics

SUVAT: $v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{1}{2}(u+v)t$

Dot product: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta = a_x b_x + a_y b_y + a_z b_z$

Cross product: $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{\mathbf{n}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$

Rocket equation: $\Delta v = u \ln\left(\frac{m_0}{m_f}\right)$

Gravitational PE: $U = -\frac{GMm}{r}$

Escape velocity: $v_e = \sqrt{\frac{2GM}{R}}$

Force from PE: $F = -\frac{dU}{dx}$

Elastic collision (1D): $v_1' = \frac{m_1 - m_2}{m_1 + m_2}v_1 + \frac{2m_2}{m_1 + m_2}v_2$

Centre of mass: $\mathbf{R}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}$

Reduced mass: $\mu = \frac{m_1 m_2}{m_1 + m_2}$

Centripetal accel: $a_c = \frac{v^2}{r} = \omega^2 r$

Parallel axis thm: $I = I_{\text{cm}} + Md^2$

Atwood machine: $a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}$

Moments of inertia (about axis through centre of mass):

Thin ring (radius R):	$I = MR^2$
Solid disk (radius R):	$I = \frac{1}{2}MR^2$
Solid sphere (radius R):	$I = \frac{2}{5}MR^2$
Hollow sphere (radius R):	$I = \frac{2}{3}MR^2$
Thin rod (length L , about centre):	$I = \frac{1}{12}ML^2$
Thin rod (length L , about end):	$I = \frac{1}{3}ML^2$

Properties of Matter

Ideal gas: $PV = nRT$

Heat capacities: $C_P - C_V = R$ (per mole)

Adiabatic: $PV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}$

Isothermal work: $W = nRT \ln\left(\frac{V_2}{V_1}\right)$

Carnot efficiency: $\eta = 1 - \frac{T_c}{T_h}$

Entropy: $dS = \frac{dQ_{\text{rev}}}{T}, \quad \Delta S = mc \ln\left(\frac{T_2}{T_1}\right), \quad \Delta S = nR \ln\left(\frac{V_2}{V_1}\right)$

Lennard-Jones: $V(r) = 4\varepsilon \left[\left(\frac{a}{r}\right)^{12} - \left(\frac{a}{r}\right)^6 \right]$

Mean KE: $\langle KE \rangle = \frac{3}{2}k_B T$

MB speeds: $c_{\text{mp}} = \sqrt{\frac{2k_B T}{m}}, \quad c_{\text{avg}} = \sqrt{\frac{8k_B T}{\pi m}}, \quad c_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$

Oscillations and Waves

SHM:	$x(t) = A \cos(\omega_0 t + \phi), \quad \omega_0 = \sqrt{k/m}$
Damped SHM:	$x(t) = A e^{-\gamma t/2} \cos(\omega_d t + \phi), \quad \omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$
Quality factor:	$Q = \omega_0/\gamma$
Forced amplitude:	$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}}$
Phase lag:	$\tan \delta = \frac{\gamma \omega}{\omega_0^2 - \omega^2}$
Pendulums:	$T_{\text{simple}} = 2\pi \sqrt{L/g}, \quad T_{\text{physical}} = 2\pi \sqrt{I/(Mgh)}$
Wave equation:	$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$
String speed:	$v = \sqrt{T/\mu}$
Sound in gas:	$v = \sqrt{\gamma P/\rho}$
Harmonic wave:	$y = A \sin(kx - \omega t)$
String impedance:	$Z = \mu v = \sqrt{\mu T}$
Reflection:	$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t = \frac{2Z_1}{Z_1 + Z_2}$
Power (string):	$\langle P \rangle = \frac{1}{2} \mu \omega^2 A^2 v$
Decibels:	$\beta = 10 \log_{10}(I/I_0), \quad I_0 = 10^{-12} \text{ W/m}^2$
Doppler:	$f' = f \frac{v \pm v_o}{v \mp v_s}$
Group velocity:	$v_g = \frac{d\omega}{dk}$
String harmonics:	$f_n = \frac{nv}{2L} \quad (n = 1, 2, 3, \dots)$
Open-closed pipe:	$f_n = \frac{nv}{4L} \quad (n = 1, 3, 5, \dots)$

Quantum Physics

Schrödinger eq: $i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$

Time evolution: $|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle, \quad U(t) = e^{-iHt/\hbar}$

Spin operators:

$$S_z = \frac{\hbar}{2} (\uparrow\uparrow\langle\uparrow - \downarrow\downarrow\langle\downarrow)$$

$$S_x = \frac{\hbar}{2} (\uparrow\uparrow\langle\downarrow + \downarrow\downarrow\langle\uparrow)$$

$$S_y = \frac{\hbar}{2} (-i \uparrow\uparrow\langle\downarrow + i \downarrow\downarrow\langle\uparrow)$$

Spin eigenstates:

$$\uparrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + \downarrow_z\rangle), \quad \downarrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - \downarrow_z\rangle)$$

$$\uparrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + i \downarrow_z\rangle), \quad \downarrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - i \downarrow_z\rangle)$$

Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Physical Constants

Speed of light	c	3.00×10^8 m/s
Planck's constant	h	6.626×10^{-34} J·s
Reduced Planck's constant	$\hbar$	1.055×10^{-34} J·s
Boltzmann constant	k_B	1.381×10^{-23} J/K
Gas constant	R	8.314 J/(mol·K)
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
Gravitational constant	G	6.674×10^{-11} N·m ² /kg ²
Acceleration due to gravity	g	9.81 m/s ²
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Elementary charge	e	1.602×10^{-19} C
Mass of Earth	$M_\oplus$	5.97×10^{24} kg
Radius of Earth	$R_\oplus$	6.37×10^6 m